SUPPORTING DOCUMENT · CREDITS

Credits and inspirations

Everything Refill is built on, borrowed from or learned from, with licences, plus the AI tools we used and who did what.

Team	Team C29 · Fynd × NMIMS Hyderabad Product & Business Simulation
Case	C29 Appointment Recovery Desk
Live app	fillthechair.vercel.app
Repository	github.com/sirgreaseball/refill
Companion file	`Refill-Citations.xlsx`: every claim we make, with its source or our test evidence
Date	18 October 2026

01 · DATA, SOFTWARE, SERVICES, FONTS

What we used

This part lists what Refill runs on and what we used to make the film, the deck and these documents. Every library is open source or free to use for a project like ours; the licence is named next to it.

The case and the data

What	Where it came from	How we used it
Case C29 brief, the three scored checks, the edge cases and the pilot path	Fynd × NMIMS Hyderabad, Use Case Book v3, p. 32 (organisers)	The product scope, the Lab and the tests follow it word for word
Starter fixture `Starter_Data/C29/fixture.json`	Organisers	Kept as scenario C01 (W1 takes S2) in `tests/core-scenarios.json`
Kesar Salon, the demo salon: staff, services, Hyderabad mid-range prices, the waiting list	Written by our team (Chirag Seervi), `data/salon/*.csv`	Synthetic. Every name is invented; no real customer data is used anywhere
Scenario tests P01–P12	Written by our team (Parth Kachhara), `tests/scenarios.json`	Run by `npm test` on every change
Market-shock rehearsal	Written by our team (Chirag Seervi), `docs/SHOCKS.md`	Click paths for late cancel, staff absence, duration change, consent withdrawal

Software inside the app

Library	What it does in Refill	Licence
React 19, React DOM	Every screen	MIT
Vite 8, @vitejs/plugin-react	Build and development server	MIT
TypeScript 6	Types across the app and tests	Apache-2.0
Vitest 5	The 51 rule and scenario tests	MIT
oxlint	Code checks	MIT
GSAP 3.15 with ScrollTrigger (GreenSock)	Scroll-driven story on the splash page	GreenSock Standard "no charge" licence
Lenis 1.3 (darkroom.engineering)	Smooth scrolling on the splash page	MIT
qrcode (node-qrcode)	QR codes for the client sign-up link	MIT

Services

Service	What it does	Plan
Supabase	PostgreSQL database (every booking rule runs here) and email-and-password sign-in	Free plan
Vercel	Hosting; every push to `main` deploys	Hobby plan

Service	What it does	Plan
GitHub	Code, issues, pull requests, CI that runs the tests on each pull request	Free
YouTube	Hosts the product film (unlisted)	Free

Fonts

Font	Designer / source	Licence
Satoshi, Clash Display	Indian Type Foundry, served by Fontshare	ITF Free Font License (free for personal and commercial use)
JetBrains Mono	JetBrains, served by Google Fonts	SIL Open Font License 1.1
Fraunces, DM Mono (team guides only)	Google Fonts	SIL Open Font License 1.1
Arial (documents and workbook)	Monotype, bundled with Windows	System font

The film, the deck and the documents

Tool	What we made with it	Licence / terms
Remotion 4	The 3:29 product film, built as React code in `video/`	Remotion License (free for individuals and small teams)
ElevenLabs	Narration: premade voice "Sarah", model eleven_multilingual_v2	Free plan; credited on the film's end card as the free plan requires
Our own scripts (`video/tools/sfx.py`, `music.py`)	All 29 sound effects and the music bed, synthesised from scratch	Ours; no third-party audio
pptxgenjs	The final deck, dark and light	MIT
docx	The Word documents, this one included	MIT
openpyxl	The citations workbook	MIT
Playwright	Screenshots of the live app for the deck and README	Apache-2.0
Pillow	Composing screenshots	MIT-CMU (HPND)
Microsoft PowerPoint, Word, Excel	Font embedding, contents pages, PDF export	Licensed Microsoft 365

02 · DESIGN, PRODUCT IDEAS, TECHNIQUES, RESEARCH

What inspired us

We studied other products and research before building. This part names each source of an idea and says what we took from it. We took ideas and patterns, never code, artwork or copy.

Design

Where in Refill	Inspired by	What we took
The Day Dial (front desk, splash, sign-in)	Chronograph watch faces	The day as a watch face: one ring per staff member, one arc per appointment, a hand at the current time. Colour only where it means something: lime filled, amber open, coral needs moving.
Splash story page	lusion.co, Apple product pages, Awwwards Sites of the Day	The feel only: a story scrubbed by the scroll and a cursor that drives the scene. Colours, layout and copy are Refill's own.
Sign-in: heading with a switch link instead of tabs	Linear, Resend	The pattern
Sign-in: show-password inside the field	Stripe, Cal.com	The pattern
Sign-in: an "or" rule before other ways in	Clerk	The pattern
Sign-in: the brand on its own panel beside the form	Stripe, Supabase	The pattern; our panel plays a refill on the Day Dial
Front-desk calendar with one column per staff member	The staff-column calendars common in salon software	The layout, with Refill's colour key (open, refilled, needs moving, booked)
Glass top bar and dock	Apple's translucent materials	A blurred bar and dock over the content
Motion and build techniques	Our own earlier projects: Threshold, Sandbox, Cascade	Patterns reused from our own code

Product ideas from other products

Idea	Seen in	What Refill does differently
A waitlist that fills gaps left by last-minute cancellations	Square Appointments, Fresha, Zenoti, Vagaro, Mangomint (citations C25–C29)	Offers go only to clients who fit the duration, the staff member's skills and hours, and their own free time, and who said yes to messages. The database enforces every rule, so no screen can skip it.
Consent recorded with time and source	India's Digital Personal Data Protection Act, 2023 and TRAI's consent register for SMS (C18, C20)	Three states, never pre-ticked: yes, no, never asked. Only yes is messaged, checked when offers go out and again at booking.
One-step opt-out	The STOP convention in text messaging	One STOP link withdraws every open offer at once.

Idea	Seen in	What Refill does differently
Late-cancellation windows and message limits	Common salon cancellation policies and anti-spam practice	Owner-set defaults: under 2 hours is late, offers held 15 minutes, at most 3 offers a client a day, no offers inside 10 minutes of the start.
An evidence log with a reason for every decision, and edge-case toggles with before/after export	Our MBA team's research on the case	Every offer, skip, booking, expiry, move and undo is logged with a rule code and exports as CSV or JSON.
Prepayment and no-show research	Treatwell, Booksy, Fresha (C11, C35, C36)	Shaped how we frame the problem. Deposits are not a Refill feature yet.

Engineering techniques

- **One slot, one booking:** a PostgreSQL row lock taken in a fixed order, plus a unique index on confirmed bookings. Both are standard PostgreSQL features.
- **The database as the only door:** the app calls database functions; tables have no direct client access. This is a common Supabase pattern.
- **Installable app:** a web app manifest and the View Transitions API, both web-platform standards.

Research we read

The outside facts behind the deck and the documents (market size, cancellation losses, software churn, prices, regulation, and the psychology of buying time) are listed one by one, with links and what each source says, in `Refill-Citations.xlsx`. Six of them need a small correction before final use; the workbook names each fix.

03 · DISCLOSURE AND CREDITS

AI tools and the team

AI tools we used

Tool	How we used it
Claude Code (Anthropic)	Dhruv's coding and writing assistant for the app, the database, the tests, the film, the deck build and these documents. Every change was reviewed, tested and committed by Dhruv; the tests in `Refill-Citations.xlsx` (Product evidence) are what prove the result.
AI coding assistants (teammates)	Used on teammates' own branches under the rules in `AGENTS.md`: which files each person may touch, and no direct pushes to `main`.
ElevenLabs	The AI voice that narrates the film.
Design guidance for the assistant	Written design rules kept in `.claude/skills/` (flagship-feel, design-taste-frontend, emil-design-eng, apple-design).

What AI did not do

It did not invent data: the demo salon is synthetic and labelled as such, and every outside number is checked against its source in the citations workbook.

It did not decide what to build: the scope comes from the case brief and the team's choices.

Who did what

Person	Team	Contribution
Vedanth Raj	B.Tech · team lead	Reliability and handoff: clean-machine setup check (`docs/SETUP_CHECK.md`), known limits, demo script, engineering presentation.
Dhruv Sonar	B.Tech	Product and code: rules engine, database and its guarantees, every screen and the design system, deployment, the film, the deck build and the team documents.
Parth Kachhara	B.Tech	Scenario tests P01–P12; real-phone testing with twelve bug reports (#9 to #20); the Lab fixes (pull request #21).
Chirag Seervi	B.Tech	Salon data and prices; the market-shock scenarios and rehearsal; the splash story page.
Parth Srivastava	MBA	Market research, segmentation and positioning, pricing hypothesis, unit economics, financial projections, go-to-market, the original deck.
Tushar Walia	MBA	Market research, segmentation and positioning, pricing hypothesis, unit economics, financial projections, go-to-market, the original deck.

Questions about any credit: open an issue at github.com/sirgreaseball/refill.